

Worksheet on Fedder's Criterion

PART 1: PROVING FEDDER

Prove each of the following results:

Lemma 1. *If there exists a map $\pi : F_*^e R \rightarrow R$ such that $\pi(F_*^e d) = 1$, and $c|d$ in R , then there exists a map sending $F_*^e c \mapsto 1$. In particular, if there is ANY surjective map $F_*^e R \rightarrow R$ then R is F -split.*

Lemma 2. *Let $\text{ev}_d : \text{Hom}_R(F_*^e R, R) \rightarrow R$ be the "evaluation at $F_*^e d$ " map, so that $\text{ev}_d(\varphi) = \varphi(F_*^e d)$. Then there exists a map $\pi \in \text{Hom}_R(F_*^e R, R)$ such that $\pi(F_*^e d) = 1$ if and only if ev_d is surjective.*

Proposition 1. *The following are equivalent for any F -finite ring R :*

- (1) R is F -split
- (2) $W^{-1}R$ is F -split for all multiplicative sets W .
- (3) $R_{\mathfrak{p}}$ is F -split for all prime ideals $\mathfrak{p}$
- (4) $R_{\mathfrak{m}}$ is F -split for all maximal ideals $\mathfrak{m}$

Corollary 1. *If R is F -finite then the locus of F -split points is open, and is specifically equal to $\text{Spec } R \setminus \mathbb{V}(\text{im } \text{ev}_1)$.*

Lemma 3. *Let S be regular, take $t \in I^{[p^e]} : I$ and let $\varphi = F_*^e t \cdot \Phi$. Show that $\Psi(\varphi)$ (i.e., φ viewed as a map on S/I) is surjective if and only if $t \notin \mathfrak{m}^{[p^e]}$.*

PART 2: USING FEDDER

You may use the following two facts without proof (see HW2 for the first one!)

Proposition 2. *If $f_1, \dots, f_t$ is a regular sequence and $I = \langle f_1, \dots, f_t \rangle$ then*

$$I^{[p^e]} : I = I^{[p^e]} + \langle (f_1 \cdots f_t)^{p^e-1} \rangle$$

Proposition 3. *If S is a graded ring with homogeneous maximal ideal $\mathfrak{m}$, then S is F -split if and only if $S_{\mathfrak{m}}$ is F -split; and Fedder's criterion holds for graded regular rings and homogeneous ideals.*

(1) Let S be a polynomial ring and I a squarefree monomial ideal. Prove that S/I is F -split?

(2) Let $S = \mathbb{F}_p[x, y, z]$ and $I = \langle x^3 + y^3 + z^3 \rangle$. For which values of p is S/I F -split?